Mock for End Term

The due date for submitting this assignment has passed.
Due on 2026-05-08, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.
2 points
Choose all the subspaces isomorphic to R^4 R4.

$W_1 = \{\begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} : a + b + c + d + e + f = 0\}$W1 = $\left\{ \begin{bmatrix} \\ \text{acebdf} \end{bmatrix} : \text{a + b + c + d + e + f = 0} \right\}$

$W_2 = \{(x, y, z, w, u) : x + y + z = w, u = 0\}$W2 = {(x, y, z, w, u): x + y + z = w, u = 0}.

$W_3 = \{\begin{bmatrix} a & b & c & d \\ e & f & g & h \end{bmatrix} : a + b + c + d = 0, e + f + g + h = 0\}$W3 = {[aebfcgdh

] : a + b + c + d = 0, e + f + g + h = 0}.

W_4 W4is the column space of the matrix $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} \\ \text{1010011011000001} \end{bmatrix}$.

W_5 W5 is the orthogonal complement of the zero subspace in R^4 R4 with respect to the standard inner product. \end{enumerate}

No, the answer is incorrect.
Score: 0
Accepted Answers:

W_4 W4is the column space of the matrix $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} \\ \text{1010011011000001} \end{bmatrix}$.

W_5 W5 is the orthogonal complement of the zero subspace in R^4 R4 with respect to the standard inner product. \end{enumerate}
2 points
Consider a basis $\{(1, 0, 1), (-1, 1, 1)\}\{(1, 0, 1), (-1, 1, 1)\}$ of a subspace WW in R^3 R3 with the standard dot product. Find the projection vector of $(1, 2, 3)(1, 2, 3)$ onto WW.

$(\frac{2}{3}, \frac{4}{3}, \frac{10}{3})(32, 34, 310)$

$(-\frac{2}{3}, -\frac{4}{3}, -\frac{10}{3})(-32, -34, -310)$

$(-\frac{2}{3}, \frac{4}{3}, -\frac{10}{3})(-32, 34, -310)$

$(\frac{2}{3}, -\frac{4}{3}, \frac{10}{3})(32, -34, 310)$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(\frac{2}{3}, \frac{4}{3}, \frac{10}{3})(32, 34, 310)$

Consider the vector space R^3 R3 with usual addition and scalar multiplication. Let S S be its subset defined as follows:

$S = \{(1, 0, 1), (0, 1, 1)\}$S = {(1, 0, 1), (0, 1, 1)}
and W_1 W1 and W_2 W2 be its vector subspaces defined as:

$W_1 = \{(x, y, z) \mid x = z, \text{ and } y = z; x, y, z \in R\}$W1 = {(x, y, z) ∣ x = z, and y = z; x, y, z ∈ R}

$W_2 = \{(x, y, z) \mid z = x + y; x, y, z \in R\}$W2 = {(x, y, z) ∣ z = x + y; x, y, z ∈ R}
Answer the following questions:

2 points
Which of the following options is (are) true?

S S is linearly independent.

S S spans the vector space R^3 R3 (with usual addition and scalar multiplication).

$Span(S)$Span(S) is a proper subspace of R^3 R3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
S S is linearly independent.
$Span(S)$Span(S) is a proper subspace of R^3 R3.

1 point
Which of the following options is (are) correct?

S S is a basis of W_1 W1.

S S is a basis of W_2 W2.

No, the answer is incorrect.
Score: 0
Accepted Answers:
S S is a basis of W_2 W2.

What is the dimension of W_1 W1?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1

1 point
What is the dimension of W_2 W2?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

Consider a map $T: V \rightarrow W$T : V → W, defined by $T(A) = trace(A)$T(A) = trace(A), where VV is the vector space of all matrices of order 2 with respect to the usual addition and scalar multiplication in matrices, and WW is RR with usual addition and scalar multiplication. Let $\beta = \{\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&1\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\begin{bmatrix}0&0\\0&1\end{bmatrix}\}$β = {[1000], [0010], [0100][0001]} be an ordered basis of VV and $\gamma = \{1\}$γ = {1} be an ordered basis of WW. Answer the following questions using the information above.

2 points
Choose the set of correct options from the following:

☐ $T(A + B) = T(A) + T(B)$T(A + B) = T(A) + T(B) for all $A, B \in V$A, B ∈ V.

☐ $T(cA) = cT(A)$T(cA) = cT(A) for all $A \in V$A ∈ V, and $c \in R$c ∈ R.

☐ TT is not a linear transformmation.

☐ $T(I) = 0$T(I) = 0, where II denotes the identity matrix of order 2.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$T(A + B) = T(A) + T(B)$T(A + B) = T(A) + T(B) for all $A, B \in V$A, B ∈ V.
$T(cA) = cT(A)$T(cA) = cT(A) for all $A \in V$A ∈ V, and $c \in R$c ∈ R.
2 points
Choose the set of correct options from the following:

○ If TT is a linear transformation and MM is the matrix representation of TT with respect to the bases ββ for the domain and γγ for the codomain, then $M = [1001]$M = [1001].

○ If TT is a linear transformation and MM is the matrix representation of TT with respect to the bases ββ for the domain and γγ for the codomain, then $M = \begin{bmatrix}1\\0\\0\\1\end{bmatrix}$M = $\begin{bmatrix}\\\\1001\end{bmatrix}$.

○ If TT is a linear transformation and MM is the matrix representation of TT with respect to the bases ββ for the domain and γγ for the codomain, then $M = [1010]$M = [1010].

○ If TT is a linear transformation and MM is the matrix representation of TT with respect to the bases ββ for the domain and γγ for the codomain, then $M = \begin{bmatrix}1&0\\0&1\end{bmatrix}$M = [1001].

No, the answer is incorrect.
Score: 0
Accepted Answers:
If TT is a linear transformation and MM is the matrix representation of TT with respect to the bases β

β for the domain and γγ for the codomain, then M = [1001]M = [1001].

3 points

Let AA and BB be $4 \times 4$4 × 4 real matrices such that rank(A) = 2rank(A) = 2 and rank(B) = 3 rank(B) = 3. Choose all the correct options from the following:

AA and BB are not equivalent.

The system $Ax = 0$Ax = 0 has infinitely many solutions and the solution space has dimension 22.

The nullity of BB is 11, and hence $Bx = 0$Bx = 0 has a unique non-trivial solution up to scalar multiples.

The columns of AA lie in a subspace of R^4R4 of dimension 22, while the columns of BB span R^4R4.

If u_1, u_2, u_3, u_4u1, u2, u3, u4 are the columns of BB, then these vectors must be linearly dependent but cannot all lie in the same plane.

No, the answer is incorrect.

Score: 0

Accepted Answers:

AA and BB are not equivalent.

The system $Ax = 0$Ax = 0 has infinitely many solutions and the solution space has dimension 22.

The nullity of BB is 11, and hence $Bx = 0$Bx = 0 has a unique non-trivial solution up to scalar multiples.

If u_1, u_2, u_3, u_4u1, u2, u3, u4 are the columns of BB, then these vectors must be linearly dependent but cannot all lie in the same plane.

3 points

Choose the correct statements for the function $f(x, y) = \begin{cases} \frac{x^2 y^2}{x^4 + y^4} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$f(x, y) = ⎩⎨⎧x4 + y4x2y20(x, y) = (0, 0)(x, y) = (0, 0)

$\lim_{(x, y) \to (0, 0)} f(x, y) = 0$(x,y)→(0,0)lim f(x, y) = 0

ff is not continuous at $(0, 0)$(0, 0).

$\lim_{(x, y) \to (0, 0)} f(x, y)$(x,y)→(0,0)lim f(x, y) does not exist.

$\lim_{(x, y) \to (1, -1)} f(x, y) = \frac{1}{2}$(x,y)→(1,−1)lim f(x, y) = 21

No, the answer is incorrect.

Score: 0

Accepted Answers:

ff is not continuous at $(0, 0)$(0, 0).

$\lim_{(x, y) \to (0, 0)} f(x, y)$(x,y)→(0,0)lim f(x, y) does not exist.

$\lim_{(x, y) \to (1, -1)} f(x, y) = \frac{1}{2}$(x,y)→(1,−1)lim f(x, y) = 21

2 points

From the options below, choose all the options containing functions defined on the main $\mathbb{R}^2 \setminus \{(0,0)\}$ for which the limit of the function at the origin along any straight through the origin exists, and the limits are all equal.

- $f(x,y) = \frac{(x-y)^2}{x^2+y^2}$
- $f(x,y) = \frac{xy}{\sqrt{x^2+y^2}}$
- $f(x,y) = (x^2+y^2)\sin\left(\frac{1}{x^2+y^2}\right)$
- $f(x,y) = \frac{xy^2}{x^2+y^4}$
- $f(x,y) = \frac{x^2-y^2}{x^2+y^2}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$f(x,y) = \frac{xy}{\sqrt{x^2+y^2}}$

$f(x,y) = (x^2+y^2)\sin\left(\frac{1}{x^2+y^2}\right)$

$f(x,y) = \frac{xy^2}{x^2+y^4}$

2 points

Consider the following functions f f defined as:

$$f(x, y, z) = xyz.\text{f(x, y, z) = xyz.}$$

Then the tangent line at the point (1,0,1) in the direction of (1,1,1) is parallel to the line

$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$x(t) = 1 + 3

√

t, y(t) = 3

√

t, z(t) = 1 + 3

√

t, u(t) = 3

√

t

$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = t\sqrt{3}$x(t) = 1 + 3

√

t, y(t) = 3

√

t, z(t) = 1 + 3

√

t, u(t) = t 3

√

$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = 1 + \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$x(t) = 1 + 3

√

t, y(t) = 1 + 3

√

t, z(t) = 1 + 3

√

t, u(t) = 3

√

t

None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$x(t) = 1 + 3

√

t, y(t) = 3

√

t, z(t) = 1 + 3

√

t, u(t) = 3

√

t

Let $f: R^2 \rightarrow R$f : R2 → R be the real-valued function defined by $f(x, y) = x^2 - xy + y^2$, for all $(x, y) \in R^2$.f(x, y) = x2 − xy + y2, for all (x, y) ∈ R2. Let $T(3, 1)$T(3, 1) denote the tangent plane to the graph of ff at the point (3, 1)(3, 1). Since $T(3, 1)$T(3, 1) is an affine subspace of R^3R3, there exists a vector $v \in R^3$v ∈ R3 and a unique subspace $U \subseteq R^3$ U ⊆ R3 such that $T(3, 1) = v + U$ = T(3, 1) = v + U = $\{v + u \mid u \epsilon U$v + u∣uϵU}.

2 points

Which of the following are possible candidates for vv

(2, 1, 2)(2, 1, 2)

(0, 7, 0)(0, 7, 0)

(1, − 2, 0)(1, −2, 0)

(2, 0, − 1)(2, 0, −1)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(2, 1, 2)(2, 1, 2)

(1, − 2, 0)(1, −2, 0)

2 points

Which of the following is a basis for UU?

$\{(1, -1, -1), (1, 3, 2)\}\{(1, -1, -1), (1, 3, 2)\}$

$\{(1, 2, 3), (-1, -1, -1)\}\{(1, 2, 3), (-1, -1, -1)\}$

$\{(0, 1, -1), (0, -1, 1)\}\{(0, 1, -1), (0, -1, 1)\}$

$\{(1, 2, 3), (0, -1, 1)\}\{(1, 2, 3), (0, -1, 1)\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(1, 2, 3), (0, -1, 1)\}\{(1, 2, 3), (0, -1, 1)\}$

Consider the function $f(x, y) = x^3 - 3x + y^3 - 3y^2$ f(x, y) = x3 − 3x + y3 − 3y2.

2 points
Which of the following points are the critical points of f f?
- (1,0)
- (0,0)
- (-1,0)
- (0,1)
- (1,2)
- (2,1)
- (-1,2)
- (2,-1)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(1,0)
(-1,0)
(1,2)
(-1,2)

2 points
Choose the correct options from the following.

(1,0) is a local minima of the function f f.

(0,0) is a local minima of the function f f.

(1,2) is a local minima of the function f f.

(-1,2) is a saddle point of the function f f.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(1,2) is a local minima of the function f f.
(-1,2) is a saddle point of the function f f.

2 points
Choose the correct options from the following.

(-1,0) is a local maxima of the function f f.

(0,1) is a local maxima of the function f f.

(1,0) is a saddle point of the function f f.

(2,1) is a local maxima of the function f f.

(2,-1) is a local maxima of the function f f.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(-1,0) is a local maxima of the function f f.
(1,0) is a saddle point of the function f f.

Let A_1 A1 and A_2 A2 be two affine spaces defined by the solution space of the system of linear equations

$$\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\\ 1 & -1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix}2\\ 2\\ 0\end{bmatrix} \quad \begin{bmatrix}1 & 0 & 1 & -1\\ 0 & 1 & 1 & 0\\ 1 & -1 & 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix}2\\ 2\\ 0\end{bmatrix}$$

and

$$\begin{bmatrix}1 & 0 & 1\\ 2 & 0 & 2\\ 3 & 0 & 3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix}2\\ 4\\ 6\end{bmatrix} \quad \begin{bmatrix}1 & 2 & 3 & 0\\ 0 & 0 & 1 & 2\\ 3 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix}2\\ 4\\ 6\end{bmatrix}$$

respectively. Let V_1 V1 and V_2 V2 be the vector subspaces of R^3 R3 corresponding to A_1 A1 and A_2 A2. Answer the following questions using the information given above.

3 points
Choose the correct options.

V_1 V1 has dimension 1, where as the dimension of V_2 V2 is 2.

V_1 V1 has dimension 2, where as the dimension of V_2 V2 is 1.

Both the V_1 V1 and V_2 V2 have dimension 3.

Both the V_1 V1 and V_2 V2 have dimension 1.

No, the answer is incorrect.
Score: 0
Accepted Answers:
V_1 V1 has dimension 1, where as the dimension of V_2 V2 is 2.

3 points
Which of the following function can represent a map from A_2 A2 to A_1 A1 correspond to the linear transformation between their corresponding subspaces whose matrix representation is given by $[-10][-1\ 0]$ with respect to the ordered bases $\{(1, 0, -1), (0, 1, 0)\}\{(1, 0, -1), (0, 1, 0)\}$ and $\{(1, 1, -1)\}$ $\{(1, 1, -1)\}$ for the domain and the codomain?

$f(2 + x, y, -x) = (2 - x, 2 + x, x)$f(2 + x, y, −x) = (2 − x, 2 + x, x).

$f(2+x, y, -x) = (2+x, 2+x, x)$f(2 + x, y, −x) = (2 + x, 2 + x, x).

$f(2+x, y, -x) = (2-x, 2-x, x)$f(2 + x, y, −x) = (2 − x, 2 − x, x).

$f(2+x, y, -x) = (2-x, 2-x, -x)$f(2 + x, y, −x) = (2 − x, 2 − x, −x).

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f(2+x, y, -x) = (2-x, 2-x, x)$f(2 + x, y, −x) = (2 − x, 2 − x, x).

Suppose the temperature (T)(T) at a point (x, y, z)(x, y, z) in R^3 R3 is given by the function

$$T(x, y, z) = e^{x+y+z}$$ T(x, y, z) = ex+y+z

Let T_1 T1, T_2 T2, and T_3 T3 be the tangent planes to the function $f(x, y) = \sqrt{9 - x^2 - y^2}$ f(x, y) = 9 − x2 − y2 at the points $(1, 2)$(1, 2), $(2, 1)$(2, 1), and $(2, 2)$(2, 2), respectively. The set of points on each plane T_1 T1, T_2 T2, and T_3 T3 form affine subspaces A_1 A1, A_2 A2, and A_3 A3, respectively, of the vector space R^3 R3, with respect to usual addition and scalar multiplication. Let V_1 V1, V_2 V2, and V_3 V3 denote the vector subspaces corresponding to the affine subspaces A_1 A1, A_2 A2, and A_3 A3, respectively. Answer the following questions using the above information.

3 points
Which of the following denotes the vector subspace $V_1 \cap V_2$ V1 ∩ V2?

$\{(x, y, -x-y) \mid x, y \in R\}$ {(x, y, −x − y) | x, y ∈ R}

$\{(-2y-2z, y, z) \mid y, z \in R\}$ {(−2y − 2z, y, z) | y, z ∈ R}

$\{(-2z, -2z, 3z) \mid z \in R\}$ {(−2z, −2z, 3z) | z ∈ R}

$\{(-2z, -z, z) \mid z \in R\}$ {(−2z, −z, z) | z ∈ R}

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(-2z, -2z, 3z) \mid z \in R\}$ {(−2z, −2z, 3z) | z ∈ R}

If mm is the dimension of V_1 V1, nn is the dimension of V_2 V2 and rr is the dimension of $V_1 \cap V_2$ V1 ∩ V2, then find the value of $m + n + r$m + n + r.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 5
2 points
3 points
Which of the following denotes the maximum rate of change of temperature (T)(T) at the point of intersection of the tangent planes T_1 T1, T_2 T2, and T_3 T3?

$\sqrt{3}e^3$ 3

$\sqrt{}$
e3

$\sqrt{3}e^{\frac{27}{5}}$ 3

$\sqrt{}$
e527

$\sqrt{3}e^{\frac{3}{5}}$ 3

$\sqrt{}$
e53

$3e^{\frac{27}{5}}$ 3e527

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\sqrt{3}e^{\frac{27}{5}}$ 3

$\sqrt{}$
e527

2 points

Which of the following options is/are true?

$T(x, y, z)$T (x, y, z) is a linear function.

$T(x, y, z)$T (x, y, z) is continuous function over R^3 R3.

$T_x(x, y, z)$Tx(x, y, z) is not continuous function over R^3 R3.

$T(cx, cy, cz) = (T(x, y, z))^c$ T (cx, cy, cz) = (T (x, y, z))c

No, the answer is incorrect.
Score: 0
Accepted Answers:
$T(x, y, z)$T (x, y, z) is continuous function over R^3 R3.
$T(cx, cy, cz) = (T(x, y, z))^c$ T (cx, cy, cz) = (T (x, y, z))c

Suppose on approaching to the point $(1, 2, 3)$(1, 2, 3), temperature is approaching to e^a ea. What is the value of aa?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 6
2 points

1 point

Let AA be an $n \times n$n × n matrix such that $A^2 = 0$A2 = 0. Choose the correct statement from the following.

Column space of AA is a subset of the null space of AA.

Null space of AA is a subset of the column space of AA.

Column space of AA is equal to the null space of AA.

The given data is insufficient to conclude the relationship between the column space of AA and null space of AA.